AC vs DC Charging

Understanding EV Charging Types

A practical guide to choosing the right charging infrastructure

KNOWLEDGE GUIDE

Document No.	**RD030**
Date	**December 2025**
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Introduction

Electric vehicle charging infrastructure falls into two categories: AC (Alternating Current) and DC (Direct Current). Understanding the difference is essential for selecting the right charging solution for your location.

The key takeaway: For the vast majority of EV charging use cases, AC charging is the right choice. DC fast charging serves a specific, limited set of applications.

How EV Charging Works

Every electric vehicle has an onboard charger (OBC) that converts AC power from the grid into DC power to charge the battery. AC chargers supply power to the vehicle's OBC, which handles the conversion. DC chargers bypass the OBC entirely and supply DC power directly to the battery at much higher rates.

AC Charging - The Primary Need

AC charging is the foundation of EV charging infrastructure. More than 90% of all EV charging happens through AC chargers - at homes, workplaces, parking lots, and commercial establishments.

- Vehicles are parked for hours: At home (8-12 hours overnight), at work (8-10 hours), at malls/restaurants (1-4 hours). AC charging is perfectly suited for these dwell times.
- Lower infrastructure cost: AC chargers require standard electrical supply (single-phase 230V or three-phase 415V). No transformer upgrades, no specialised equipment.
- Lower hardware cost: AC chargers are significantly more affordable than DC chargers.
- Grid-friendly: AC chargers draw moderate power (3.3 kW to 22 kW), which does not strain the local electrical grid.
- Simpler installation: Wall-mounted, compact, and easy to install by any certified electrician.
- Better for battery health: Slower, steady charging is better for long-term battery life compared to repeated fast charging.
- Universal compatibility: All EVs support AC charging via Type 2 or 3-PIN connectors.

DC Fast Charging - Specific Use Cases Only

DC fast charging (25 kW to 350+ kW) is designed for a narrow set of use cases where speed is critical and vehicles cannot wait:

- Highway corridors and expressways: For long-distance travellers who need a quick top-up (20-40 minutes) before continuing their journey
- Fuel stations: Traditional petrol/diesel stations adding EV charging to serve transit traffic
- Charge Point Operators (CPOs): Operators building public fast-charging networks along transit routes
- Utility companies and power providers: Grid operators deploying public charging infrastructure

AC vs DC Comparison

Parameter	AC Charging	DC Fast Charging
Power Range	3.3 kW to 22 kW	25 kW to 350+ kW
Charging Time	1.5 to 14 hours	20 to 60 minutes
Hardware Cost	Low	Very High (10x to 50x)
Installation Cost	Low	High (transformer, civil work)
Grid Impact	Minimal	Significant
Monthly Operating Cost	Low	High

Best For	Destinations (home, work, hotel, hospital, restaurant, mall, office)	Transit (highways, fuel stations)
Battery Impact	Gentle, preserves battery life	Degrades battery faster with repeated use
Space Required	Minimal (wall mount)	Significant (standalone + transformer)
Electrical Supply	Standard residential/commercial	Dedicated HT/LT connection often required

What Your Location Needs

Hotels, hospitals, offices, malls, restaurants, apartments, and similar destination locations only need AC charging. DC is strictly for transit corridors.

Location Type	Charging Need	Recommended Solution
Home / Villa	Overnight charging	AC 3.3 kW or 7.4 kW (POWERPOD Lite)
Apartment Complex	Resident + visitor charging	AC 7.4 kW or 22 kW (POWERPOD Go/City)
Office / Workplace	Employee daytime charging	AC 7.4 kW or 22 kW (POWERPOD Go)
Large Campus	Distributed employee charging	AC 7.4 kW (POWERPOD Go)
Hotel / Resort	Guest overnight charging	AC 7.4 kW or 22 kW (POWERPOD Go)
Hospital	Staff and visitor parking	AC 7.4 kW or 22 kW (POWERPOD Go)
Restaurant / Cafe	Dine-in parking (1-3 hrs)	AC 7.4 kW or 22 kW (POWERPOD Go)
Shopping Mall	Shopper parking (1-4 hrs)	AC 7.4 kW or 22 kW (POWERPOD Go/City)
Fleet Depot	Overnight fleet charging	AC 7.4 kW or 22 kW (POWERPOD Go)
Highway / Expressway	Transit quick charge	DC 25 kW+ (not POWERPOD range)
Fuel Station (Highway)	Transit quick charge	DC 25 kW+ (not POWERPOD range)
Fuel Station (City)	Mix of local + transit	AC 22 kW (POWERPOD City) + DC optional

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